

EEF Meeting Minutes, Thursday September 26, 2024

Erlang Ecosystem Foundation, Minutes of a Regular Meeting of the Board of Directors
Meeting #106, September 26, 2024

A regular meeting of the Erlang Ecosystem Foundation was held over group conference call (Zoom). The meeting began at 16:06 UTC.

Attendance

The following members of the board of directors were present (10/11): Francesco Cesarini, Maxim Fedorov, Kiko Fernandez-Reyes, Kenneth Lundin, Amos King, Ben Marx, Miriam Pena Villanueva, Peer Stritzinger, Sebastian Strollo, and Alistair Woodman.

The following members of the board of directors were absent: Bryan Paxton.

Also present was Paula Calgaro from the marketing team

Agenda Items

1. Approval of minutes from previous meeting.

RESOLVED, that the Erlang Ecosystem Foundation approve the minutes as presented as representing a true and accurate record of the board meeting #104 held on September 12, 2024. Approved by 7 votes, three abstained as they were not present at the last meeting (Amos, Francesco, Ben)

2. Intermediate actions: none
3. Unanimous consent votes: none scheduled.
4. No financial update. Membership update: Basic 1633, Annual supporting 128, Lifetime supporting 23, Managing and contributing 14.
5. Amos reported that Raúl Chouza has made good progress on the Gleam registry library (that we approved a stipend for back in March). He will make a video and blog post presenting his work.
6. Event reports

1. The thinking Elixir podcast that Alistair was a guest on is out - useful form of outreach!
2. Linux Foundation Open Source Summit (OpenSSF) in Vienna: Alistair was interviewed by Linux Foundation hosts, met with a lot of people.
3. EuroBSDCon in Dublin: Alistair and Peer were there, good chats around working with stipends, interesting talks (e.g. "how to talk to the EU")
4. NSSS (Nordic Software Security Summit) in Stockholm: Kiko and Alistair gave a talk which received positive feedback. Lots of useful connections.
7. Organizations to engage with, for example: Linux Foundation (we are now members), OpenSSF (in progress), OpenChain, FreeBSD Foundation, Open Source Business Alliance (OSBA, mostly German), Eclipse Foundation (and ORCWG)... In general we should define a streamlined process, need to decide for each: point of contact, does the board need to approve a membership in a new organization, perhaps develop a template with questions for each org, to be presented to the board. Ongoing work.
8. Stipend requests
 1. Dániel Horpácsi stipend request to develop a formal semantics for core Erlang: Peer presented the request, input from OTP team is that core Erlang may go away in the future, and that maybe they should target a subset of Erlang instead. Dániel and the team has already built some previous research and tooling on core Erlang, and asked Peer (as sherpa) to put the proposal as presented to a vote. Before the motion could be put to a vote Ben and Miriam had to leave the meeting.

RESOLVED, that the Erlang Ecosystem Foundation approve the \$6,000 stipend request from Dániel Horpácsi to develop a formal semantics for core Erlang. Vote count: 6 yes, 2 no (Yes: Amos, Kiko, Peer, Francesco, Sebastian, Alistair; No: Kenneth, Maxim).

9. Alistair declared the meeting adjourned at 17:06 UTC.

Dated September 26, 2024

Secretary of the meeting: _____ Sebastian Strollo

Attachments: Stipend proposal from Dániel Horpácsi (2 pages)

Stipend Proposal

Stipend Type : Development Work

First name : Dániel

Last Name : Horpácsi

Nick Name : Daniel

Organization Name : Eötvös Loránd University

Country : Hungary

Website : <https://inf.elte.hu/en/>

Intended product of project :

We want to enhance our formal semantics for Core Erlang and, based on the semantics, produce a tool that carries out formally based evaluation and lightweight verification. Formal semantics is a mathematical description of the programming language, defining programs' behaviour in an abstract, general way. In particular, formal semantics allows one to inspect all potential executions or even reason about them with mathematical precision, enabling formal reasoning about program properties. Our team has developed formal semantics for Core Erlang, a representative subset of the Erlang programming language; this definition is based on all available sources, including the language specification, the language documentation, previous research and the reference implementation. Amongst others, our implementation, written in the Coq proof assistant, proves that all executions respect the principle of signal ordering guarantee. Based on this semantics definition, we can develop various trustworthy programming language tools, the first of which will be a formally-based evaluator. This tool will allow Erlang programmers to inspect program behaviour in the formally defined program semantics, and check program properties over multiple executions, while the results will be correct by construction since the semantics is faithful to the implementation.

Purpose

Erlang/OTP is the Erlang programming language's reference implementation, defining programs' behaviour (via Core Erlang and BEAM) in concrete executions. Program behaviour is explored by evaluating example programs, and this supports program testing and validating program properties. While testing side-effecting sequential programs can be challenging, testing concurrent programs is significantly harder because of the various ways that scheduling can combine the behaviours of individual processes. Using the formal semantics for exploring program executions we can facilitate formally based execution and show simple properties of executions, for which the underlying system can provide formal proofs. This is a nice exploitation of the formal

semantics definition, and all users of the Erlang ecosystem can benefit from it by using it as a point of reference and a preliminary prover for concrete executions.

Development Plan

Our current formal semantics is a small-step, frame stack style definition - it is not inherently executable but can be turned into an interpreter (and later into a model checker) by a series of clearly defined technical steps. We propose a project that takes the current formalisation, reshapes the definition to support automatic execution, and extracts a tool from it that can be used for simple reasoning about program behaviour even in the presence of concurrency and non-determinism. In stages, the project team will do the following tasks, each taking about one month: - Investigate the automatic execution of the frame stack style semantics of the Core Erlang language; revise the definition if needed. - Investigate the extraction of the semantics (along with the execution tactics, if any) into a functional program that serves as the basis of the formally-based evaluator (making sure the conversion between Erlang and Core Erlang does not hinder practical usage of the solution). - Develop a front end to the formal semantics, which displays information about potential execution paths and execution results in an understandable format. - Add support for checking simple specifications (e.g., in terms of simple assertions and post-conditions) and define some case studies that demonstrate the approach. - Evaluate the effectiveness of the system on a range of sequential and concurrent Erlang programs. - Finalise documentation and write-up.

Dissemination : We will conduct the work so that the results can be presented at a developer conference/meetup and published in the peer-reviewed literature.

Delivery Method : The deliverables of the project will be: - An executable tool that allows users to use the formally-based evaluator to run their programs, inspect potential executions and check simple properties. - A write-up of the results in a form that can be presented to a developer conference and published in the peer-reviewed literature. - The code itself will be available through a suitable open-source GitHub repository, along with user documentation.

Stipend Amount (USD) : 6000